Name	Subject Class	Class	Index Number
	2BI		

ANGLO-CHINESE JUNIOR COLLEGE
H2 Biology Preliminary Examinations 2021

H2 BIOLOGY **9744/04**
Practical
12 August 2021
2 hours and 30 minutes

READ THESE INSTRUCTIONS FIRST
Write your name, index number, class, shift and laboratory on this Question Paper.

Write in dark blue or black pen.
You may use a HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Shift
Laboratory

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate. You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
1	/ 28
2	/ 27
Total	**55**

This question paper consists of 16 printed pages.

Answer **all** questions.

1 Yeast cells contain enzymes which catalyse the breakdown of glucose to produce ethanol and carbon dioxide.

These products change the environment of the yeast cells and can affect their metabolic activity and survival.

The carbon dioxide produced dissolves in the aqueous medium to form a weak acid. Hence the more carbon dioxide released, the more acid will be formed.

(a) Suggest why the presence of ethanol, a small organic molecule, can affect the survival of yeast cells.

....

....

....

.... [2]

You are required to investigate the effect of different concentrations of ethanol on the metabolic activity of yeast cells by measuring the change in pH, using Universal Indicator paper strips.

You are provided with:

- at least 60 cm^3 yeast suspension, labelled **Y**
- at least 15 cm^3 ethanol, labelled **E**
- at least 60 cm^3 20% glucose solution, labelled **G**
- distilled water

Ethanol is harmful and highly flammable. If any comes into contact with your skin, wash immediately under cold water.

It is recommended that you should wear eye protection.

Keep the ethanol covered when you are not using it.

It is important to determine the pH of the glucose solution and the ethanol before starting the investigation.

Place a strip of the Universal Indicator paper on a white tile.
Using a clean Pasteur pipette, place a drop of glucose solution onto the paper.
Use the colour chart to identify the pH.
Repeat the steps above, after cleaning the Pasteur pipette, to find the pH of ethanol.

(b) (i) Record the pH for both the glucose solution and ethanol in the table below.

Solution	pH
Glucose	
Ethanol	

[1]

Clean and dry the white tile before proceeding with the rest of the procedure.

You are going to vary the independent variable, the concentration of ethanol.

Table 1.1 shows how to prepare two of the concentrations you should use. These two concentrations are the lowest and highest concentrations that need to be prepared.

Table 1.1

percentage concentration of ethanol	volume of ethanol / cm^3	volume of distilled water / cm^3
0	0.0	10.0
40	4.0	6.0

(ii) Using the information in Table 1.1, decide which other ethanol concentrations to prepare and complete Table 1.2. Include the ethanol concentrations from Table 1.1.

Table 1.2

tube number	percentage concentration of ethanol	volume of ethanol / cm^3	volume of distilled water / cm^3

[3]

1 Label 5 boiling tubes **1** to **5**.

2 **Adding the water before the ethanol**, use the syringes provided to make up the ethanol concentrations in the correct boiling tubes.

3 Use a clean 10 cm^3 syringe to put 10 cm^3 of the yeast suspension into each boiling tube.

4 Use the beaker provided to make a water bath with warm water between 45 °C and 50 °C.

5 Shake the tubes carefully to thoroughly mix the ethanol and water.

6 Place the tubes into the warm water and leave them for **at least** 5 minutes.

7 Use the marker pen provided to label the white tile as shown in Fig. 1.1.

8 Arrange two rows of the Universal Indicator paper strips on the white tile as shown in Fig. 1.2.

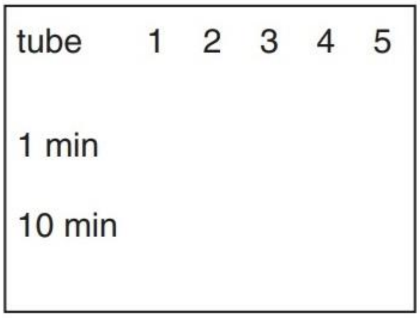

Fig. 1.1

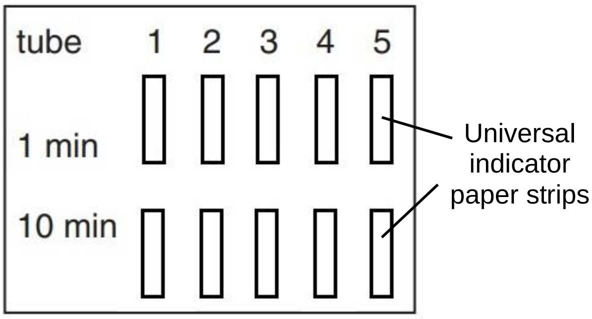

Fig. 1.2

9 After the tubes have been in the water bath for at least five minutes, start a stopwatch.

10 Use a clean 10 cm^3 syringe to put 10 cm^3 of the glucose solution into each tube starting with tube 1.

Each time shake the tube well and return it to the warm water bath.

11 When the stopwatch shows one minute, use the glass rod to remove a drop from the contents of tube 1 and place the drop onto the correct Universal Indicator paper strip.

12 Clean the glass rod and use it to remove a drop as described in step **11** from the other four tubes.

13 When the stopwatch shows 10 minutes, use the glass rod to take further drops as described in step **11**.

(iii) Prepare the space below and record the pH corresponding to the colour of each Universal Indicator paper strip.

[4]

(c) (i) Identify **two** significant sources of error in this investigation.

....

....

....

.... [2]

(ii) Syringes were used to measure the volumes of ethanol and distilled water.

State the volume of the smallest division on the syringe ....

State the degree of uncertainty .... [1]

(d) Using the relevant information provided, design an experiment that will allow you to determine the lowest concentration of ethanol that will result in zero metabolic activity in yeast cells. You should use a method, other than the one described above.

You should make use of the following apparatus:

- 500 cm^3 glass beaker
- Boiling tube
- Delivery tube

You may use the space given below to draw your proposed experimental set-up if required.

[5]

(e) In a similar investigation, a student made up different masses of dried yeast into a suspension with 10% glucose solution and measured the percentage of light absorbed by the suspension. The results of the student's investigation are shown in Table 1.3.

Table 1.3

Mass of dried yeast / g 100 cm^{-3} glucose solution	Mean percentage absorbance
1.00	9
1.50	17
2.00	39
2.25	53
3.00	94

(i) Plot a graph of the data in Table 1.3.

[4]

The student then investigated the growth of yeast. The dry mass of 1.50 g of yeast in 100 cm^3 glucose solution has an absorbance of 17%. He measured the percentage absorbance after one day and after three days. The first result is shown in Table 1.4.

Table 1.4

time/ days	mean percentage absorbance	Mass of yeast / g $100cm^{-3}$ glucose solution
0	17	1.50
1	60	
3	25	

(ii) Complete Table 1.4 using the graph. [1]

(iii) Show clearly **on the graph** how you obtained the mass of dried yeast after one day. Put your answer on the graph that you have plotted. [1]

(iv) Based on the student's observations, he made the hypothesis that:

There will be a higher mass of yeast in the culture after two days compared to day 1.

State whether this hypothesis is supported by the student's results. Explain your answer.

....

....

....

.... [2]

(v) Discuss whether the above method used by the student in **(e)** provides an accurate measurement of the growth of yeast.

[2]

[Total: 28]

2 In this section, you will require access to a microscope.

Specimen **F** is the fruit of a common plant. Use the knife provided to cut the fruit in half as shown in Fig. 2.1.

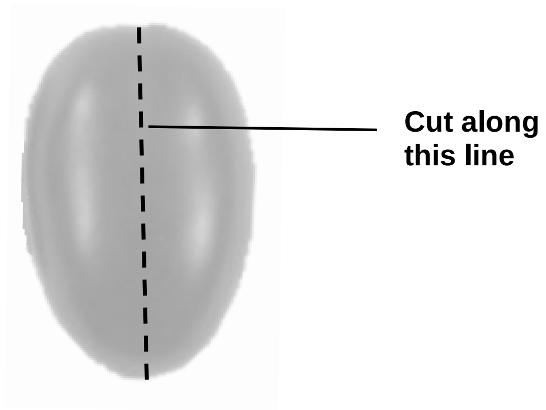

Fig. 2.1

(a) (i) Use the space provided to draw a large plan diagram of the cut surface of the fruit. Labels are **not** required.

[4]

(ii) Take the fruit and scrape some flesh from under the skin.

Place a small quantity of the flesh on a microscope slide. Add a drop of water.

Add a coverslip to the slide, lightly squashing the tissue, taking care to trap as few air bubbles as possible.

Use a paper towel to absorb any excess liquid.

Use your microscope, with an appropriate objective lens, to observe the cells.

Make a large labelled drawing of **three** cells.

[4]

(iii) An eyepiece graticule scale can be used to measure the length of cells. To obtain an actual length, the graticule scale must be calibrated against a stage micrometer.

However, to obtain values for calculating a ratio, it is **not** necessary to calibrate the eyepiece graticule scale.

Observe **the cells** of specimen **F** using the **low power** (x10) objective lens of your microscope.

Use the eyepiece graticule scale to find

- the width of the largest organelle in the cell
- the width of the cell

State the ratio between the width of the largest organelle and the cell.

You may lose marks if you do not show all the steps in finding the ratio.

Ratio .... [3]

(iv) Changes occur to specimen **F** as it undergoes the ripening process.

A student suggested the hypothesis:

A ripe specimen **F** contains more reducing sugars than the unripe version.

Explain why the student suggested this hypothesis.

....

.... [1]

(v) You are provided with

- 5 cm^3 of an extract made from unripe specimen **F**, in a vial labelled **U**
- 5 cm^3 of an extract made from ripe specimen **F**, in a vial labelled **R**
- 5 cm^3 of Benedict's solution, in a vial labelled **B**

Using the test tubes and the other apparatus available, plan **and** carry out a method to obtain results to test the student's hypothesis.

Outline the steps in your method that you used to collect results.

....

....

....

....

....

.... [3]

(vi) Record your results in a suitable format in the space provided.

[3]

(b) **S1** is a slide of a stained transverse section through a stem of a water plant.

You are not expected to be familiar with this specimen.

Examine **S1** carefully using the **low power** objective lens of your microscope.

Fig. 2.2 is a photomicrograph of a stained transverse section through a different plant stem.

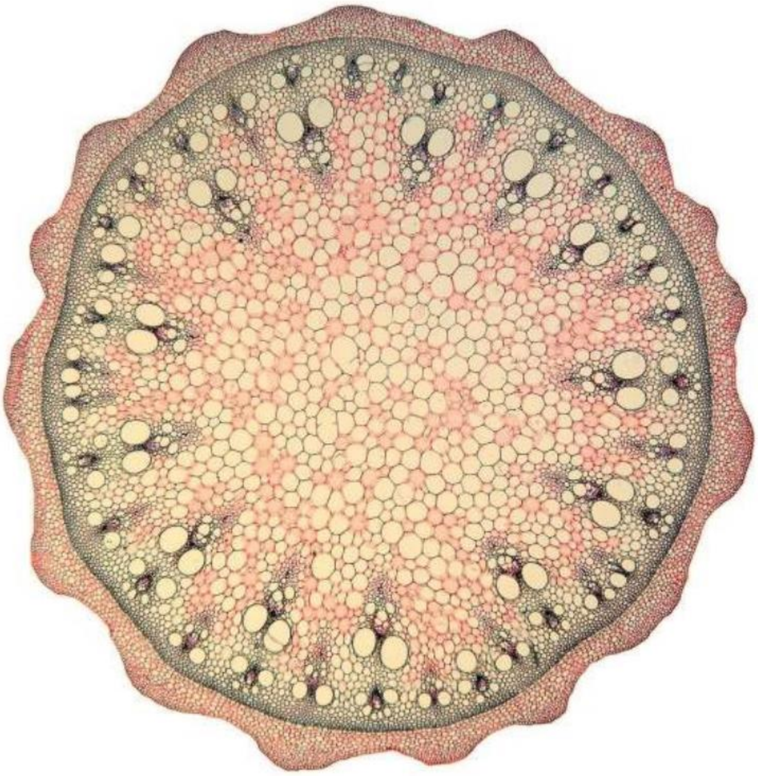

Fig. 2.2

Prepare the space below so that it is suitable for you to record observable differences between the specimen on **S1** and **Fig. 2.2**. Record your observations in the space that you have prepared.

[3]

(c) Fig. 2.3 is a photomicrograph of a stained transverse section of part of a leaf from another plant. A grid has been placed over the photomicrograph to help you answer the question. Each square is 1 cm^2.

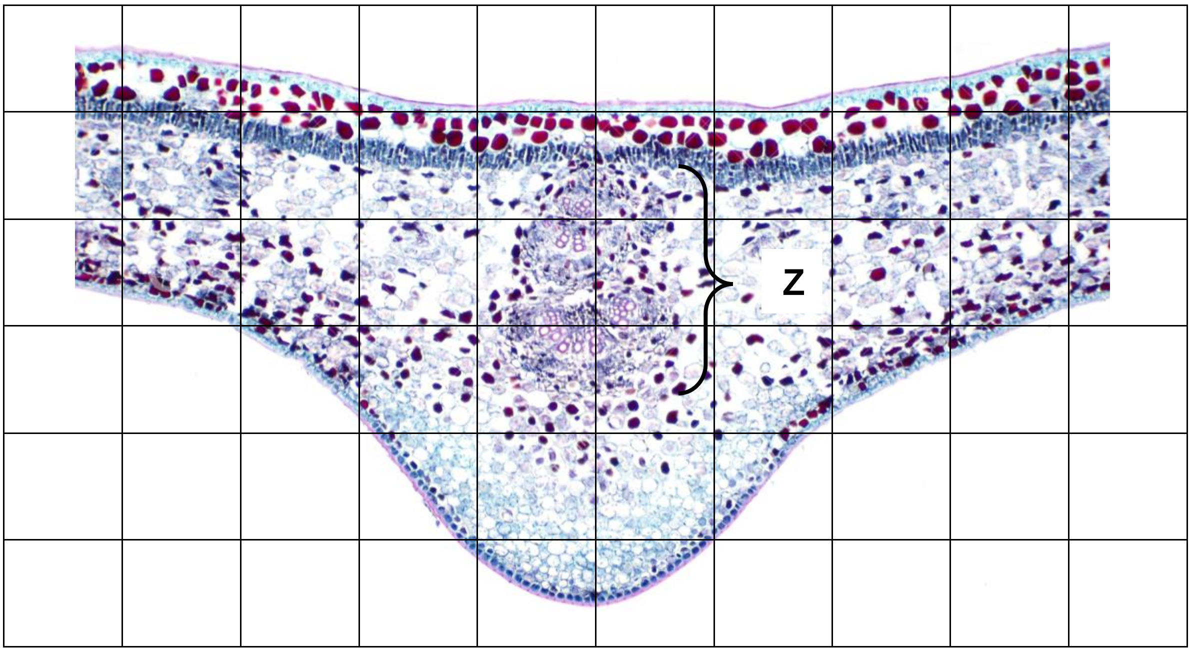

Fig. 2.3

(i) Describe how you will use the grid to find the total area of the part of leaf shown in Fig. 2.3.

....

.... [1]

(ii) Use the procedure you have described in **(c)(i)** to find:

the total area of the part of the leaf shown in Fig. 2.3 .... cm^2

the area of the leaf section occupied by the vascular bundle, labelled **Z** .... cm^2

[2]

(iii) Calculate the percentage of the part of the leaf shown in Fig. 2.3 that is occupied by the vascular bundle, labelled **Z**.

Show all the steps of your working.

.... % [2]

(iv) Suggest how you could modify the procedure you have used in **(c)(i)** to give a more accurate estimate of the area of the leaf.

....

.... [1]

[Total: 27]